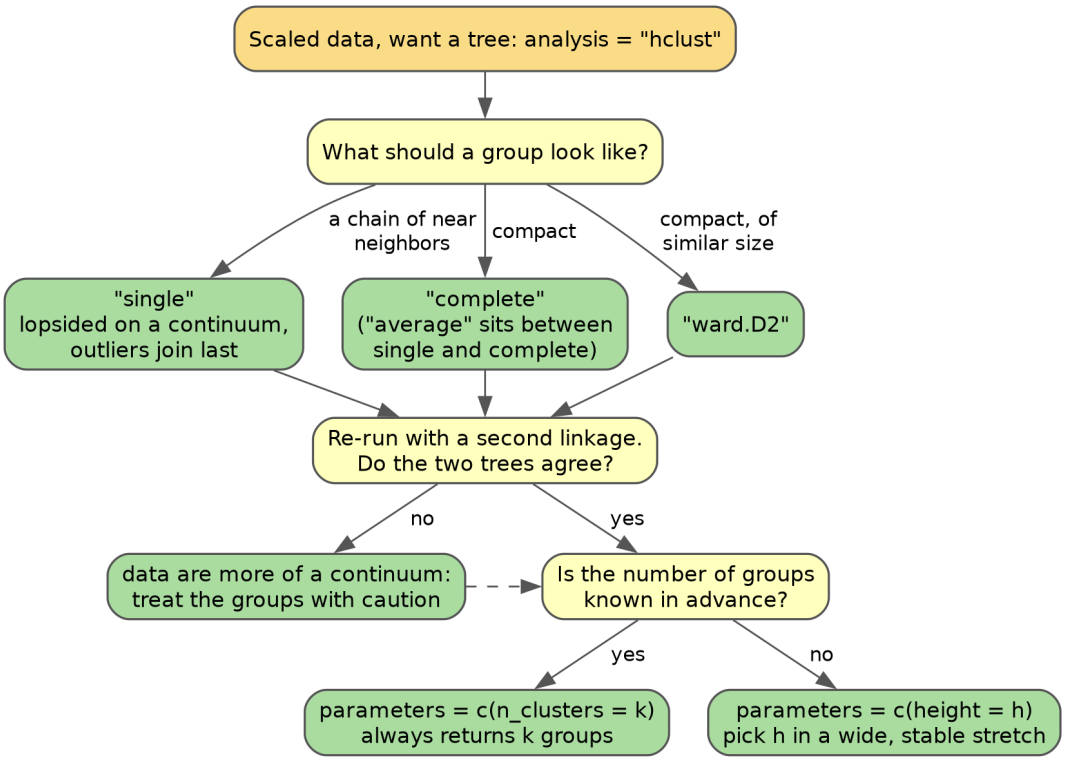
# hierarchical clustering: summary

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Hierarchical clustering merges the two closest groups, again and again, until every sample is connected by one tree structure. **The linkage rule decides what "closest" means for a group**, and different rules build different trees from the same distances. **A tree is read across, not down** (assuming it's drawn with the root on the left and leaves on the right): how far out from the tips two paths meet shows how different two samples are, and adjacent tips are not necessarily similar. **Cutting the tree turns it into groups**: choosing a height as the basis for the cut lets the data drive how many clusters are created, whereas setting a number of groups will force the tree to be cut into that number of clusters.

| Function or setting | What it does | What to set or watch |
|---|---|---|
| `runMatrixAnalysis(analysis = "hclust")` | builds the tree from the distances | `scale_variance = TRUE`, `agglomeration_method`, `parameters` |
| `agglomeration_method = "single"` | joins on the closest members | chains a continuum into a lopsided tree |
| `agglomeration_method = "complete"` | joins on the furthest members | compact groups |
| `agglomeration_method = "average"` | joins on the average distance | sits between single and complete |
| `agglomeration_method = "ward.D2"` | adds least spread within groups | compact groups of similar size |
| `parameters = c(height = 5)` | cuts at a height, adds a `cluster` column | the data decide how many groups |
| `parameters = c(n_clusters = 4)` | cuts into a set number of groups | always returns that number |
| `ggtree()` + `geom_tiplab()`, `geom_tippoint()` | draws the tree and labels the tips | `aes(label = lake)`, `aes(fill = park)`, `align = TRUE` |
| `geom_cut(height = 5)` or `geom_cut(k = 4)` | draws the cut on the tree | the same height as `parameters`, never `geom_vline()` |
| `filter(isTip)` + `aes(y = y)` in `geom_tile()` | heat map in the tree's tip order | same `scale_y_continuous(limits = ...)` as the tree, then `plot_grid()` |